

Studying the Putnam

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Problem Solving Seminar

Department of Mathematics
California State University Fullerton

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What is the Putnam Exam?

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- afternoon session: 1PM-4PM (6 more problems)

Why take the Putnam?

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- bragging rights!

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- not just for supergeniuses or the criminally insane

Getting Putnam practice

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- MAA Mathematics Magazine

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- Kiran Kedlaya's website

Putnam 2016 A1:

Find the smallest integer j such that for every polynomial $p(x)$ with integer coefficients

$p^{(j)}(k)$ is divisible by 2016 for all integers k .

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- try to explore *specific cases*

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Topics: polynomials, divisibility

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- **Fundamental Theorem of Arithmetic:**
every positive integer may be written as a product of primes

$$n = p_1 p_2 \dots p_r$$

moreover, given another prime factorization

$$n = q_1 q_2 \dots q_s$$

then $r = s$ and the p_j 's and q_j 's are the same, after reordering

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Also, polynomials are linear combinations of **monomials**: x^n

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Also, polynomials are linear combinations of **monomials**: x^n

$$f(x) = x$$

$$f'(k) = 1$$

$$f''(k) = 0$$

$$f'''(k) = 0$$

$$f''''(k) = 0$$

$$f(x) = x^2$$

$$f'(k) = 2k$$

$$f''(k) = 2$$

$$f'''(k) = 0$$

$$f''''(k) = 0$$

$$f(x) = x^3$$

$$f'(k) = 3k^2$$

$$f''(k) = 6k$$

$$f'''(k) = 6$$

$$f''''(k) = 0$$

$$f(x) = x^4$$

$$f'(k) = 4k^3$$

$$f''(k) = 12k^2$$

$$f'''(k) = 24k$$

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$$f(x) = x^{29}$$

$$f'(k) = 29k^{28}$$

$$f''(k) = 29 \cdot 28k^{27}$$

$$f'''(k) = 29 \cdot 28 \cdot 27k^{26}$$

$$f''''(k) = 29 \cdot 28 \cdot 27 \cdot 26k^{25}$$

How long until divisible by 2016?

We need five 2's, two 3's and a 7:

$$f^{(5)}(k) = 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25k^{24}$$

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Go one more to get a 2:

$$f^{(6)}(k) = 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25 \cdot 24k^{23}$$

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$$f^{(8)}(k) = 55 \cdot 54 \cdot 53 \cdot 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 k^{47}$$

We guess that the **answer is 8**

Writing up the solution

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- Good writing is important!
- ALWAYS write the answer first
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- Emphasize any theorems you are using
- Emphasize key points
- Break up your solution into manageable pieces with Lemmas
- Double-check you answered the original question!

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Now, for $f(x) = x^{55}$ we can calculate

$$f^{(7)}(1) = 55 \cdot 54 \cdot 53 \cdot 52 \cdot 51 \cdot 50 \cdot 49,$$

which is not divisible by 2016, since it isn't divisible by 2^5 .

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To prove that 8 is enough, we'll prove two Lemmas.

Lemma: If n and k are integers with $k > 0$, then one of the integers

$$\{n - 0, n - 1, n - 2, \dots, n - k + 1\}$$

must be a multiple of k .

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Lemma: If n and k are integers with $k > 0$, then one of the integers

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must be a multiple of k . **Proof:**

The value of n modulo k , ie. the remainder of n/k , is an integer r with $0 \leq r \leq k - 1$.

Therefore $n - r$ will be 0 modulo k , ie. it will be a multiple of k .

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Lemma: If $n \geq 0$ is an integer and $f(x) = x^n$, then $f^{(j)}(k)$ is divisible by 2016, for any integers j and k with $j \geq 8$.

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Proof:

$$f^{(8)}(x) = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)x^{n-8},$$

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$f^{(8)}(x) = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)x^{n-8}$,
and therefore for any integers j and k with $j \geq 8$, $f^{(j)}(k)$ will be a multiple of

$$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7).$$

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$N = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)(n-7)$. Consider the numbers $\{n, n-1, n-2, n-3, n-4, n-5, n-6, n-7\}$. At least four of the above numbers must be even, and therefore divisible by 2. Actually, one must be a multiple of 4! Likewise, at least 2 will be multiples of 3. Finally, one of the numbers will be a multiple of 7. Hence N and $f^{(j)}(k)$ is divisible by 2016.

Writing up the solution

To finish our solution, note that any polynomial will be of the form

$$f(x) = a_0 + a_1x + a_2x^2 + a_nx^n$$

for some integers $a_0, \dots, a_n$.

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By our Lemma, when we take the j 'th derivative for $j \geq 8$, each of the powers of x becomes a multiple of 2016.

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By our Lemma, when we take the j 'th derivative for $j \geq 8$, each of the powers of x becomes a multiple of 2016.

This makes $f^{(j)}(k)$ a linear combination of multiples of 2016, and therefore also a multiple of 2016.

Try it yourself!:

Find the smallest integer j such that for every polynomial $p(x)$ with integer coefficients

$p^{(j)}(k)$ is divisible by 2024 for all integers k .